

# Understanding the impacts of extreme temperature and humidity compounds on winter wheat traits in China

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## ABSTRACT

Climate change has heightened the occurrence of extreme compound events, yet their impacts on crop yield traits and the variations in yield sensitivity among different varieties and management practices remain elusive. Utilizing long-term (1981–2010) wheat trait data of 85 stations administered by the China Meteorological Center (CMC) for various wheat varieties and sowing dates, we employed a linear mixed models to explore the effects of extreme atmospheric temperature and humidity compound events on wheat. Our findings revealed that the yield loss ( $23.24 \text{ kg ha}^{-1} \text{ dec}^{-1}$ ) was primarily driven by an increasing trend ( $0.25 \sim 1.50 \text{ day dec}^{-1}$ ) in hot-dry days (HDD) during 1981–2010. Additionally, cold-wet days (CWD:  $-1.75 \%$  per standard unit) and hot-dry days (HDD:  $-1.90 \%$  per standard unit) showed similar adverse effects on yield. This similarity is primarily due to their comparable negative impacts on grain growth, with reductions in grain weight ( $-1.82 \%$  for CWD and  $-1.65 \%$  for HDD) and grain filling rate ( $-1.56 \%$  for CWD and  $-1.27 \%$  for HDD) during the heading and maturity stages of wheat. However, CWD dominated the unfavorable effects on grain number ( $-1.18 \% \sim -0.22 \%$ ). Canopy height and ear density were more susceptible to early extreme climatic factors. In addition, yield losses under early maturing varieties and late sowing measures were less sensitive to extreme climates. Our findings emphasize the negative impacts on yield compositions, but different changes in wheat yield under different varieties and sowing caused by CWD and HDD. Our findings could assist modelers in comprehending how extreme weather affects agricultural production, particularly in terms of variations in crop trait responses.

## 1. Introduction

In the context of global climate change, there is not only an increase in the frequency of extreme climate but also a growing occurrence of compound extreme events (Ding and Ke, 2015; Otto et al., 2017; Ridder et al., 2022; Zhou and Liu, 2018). Compound extreme events involve the simultaneous spatial occurrence or consecutive temporal events of two or more extreme events. They also encompass situations where the combined impact of two extreme or non-extreme events surpasses that of individual events (Field, 2012). The substantial and unpredictable consequences of these extreme compound events on global socio-economic and ecological systems are raising significant concerns. Within these events, compound temperature and moisture or humidity extremes impact crop growth through three pathways:

crop-physiological interactions, heat-moisture interactions, and crop-atmosphere interactions, resulting in adverse effects and high uncertainty in agricultural production systems (Lesk et al., 2022). Despite the notable focus on high temperatures, soil moisture, or precipitation in past research (Cohen et al., 2021; Lesk et al., 2021; Proctor et al., 2022), there has been a predominant emphasis on the impacts of temperature-moisture couplings on agriculture. This focus has often overlooked the compound effects of extreme atmospheric humidity and temperatures.

The temperature and atmospheric humidity, crucial factors in the soil-plant-atmosphere continuum, influence key crop processes like photosynthesis, respiration, and transpiration by regulating stomatal opening and closing (Medrano et al., 2002; Scafaro et al., 2021). Specifically, elevated atmospheric humidity could lead to a decrease in

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stomatal conductance, resulting in reduced photosynthetic and transpiration rates, increased respiratory intensity, and ultimately leading to decreased crop yields (Bunce, 2000; Mott and Peak, 2010). Cold and wet atmospheric conditions above the wheat canopy during the flowering period, as well as dry and hot air, can lead to unsuccessful pollination and a decline in crop grain numbers (Coast et al., 2016; Fu et al., 2023). Furthermore, cold and wet air in the later spring can increase the vulnerability of crops to diseases, such as rust and powdery mildew, ultimately resulting in diminished crop yields (El Jarroudi et al., 2017).

Due to uncertainties and a lack of robust simulation processes for extreme events in crop models, earlier studies heavily relied on county-level yield data and utilized panel statistical models to assess the impact of extreme compound events on crop yields (Ben-Ari et al., 2018; Hamed et al., 2021; Zhao et al., 2022). However, the absence of long-term observational data on yield traits has hindered these assessments, preventing them from capturing the sensitivity of yield traits to various extreme climate indices across different crop varieties and management practices. This limitation constrains their ability to guide breeding and field management strategies under climate change.

Wheat, which supplies 20 % of global protein and calorie needs, is a crucial staple food for 2.5 billion people in 89 countries (Shiferaw et al., 2013). While winter wheat thrives in cold conditions, its extended growth period makes it particularly vulnerable to extreme climates. China, a major global wheat producer, accounts for 17.8 % of the world's total wheat production (FAO, 2016), with most cultivation concentrated in the Huang-Huai-Hai Plain. The climate in this region is affected by the Siberian cold air mass and the Western Pacific subtropical high, making wheat growth susceptible to extreme temperature and humidity events (both hot/cold and wet/dry). Notably, quantitative research in China on the impact of these compound extreme temperature and humidity events on wheat yield traits, including different varieties and management practices, remains limited.

Therefore, utilizing data from long-term (1981–2010) wheat trait observation experiments and corresponding meteorological records from specific stations, this study aims to accomplish the following objectives: (1) analyze the spatiotemporal trends of extreme climate indices during the wheat growth period and identify their driving forces on yield trends, (2) quantify the sensitivity of various yield traits to compound events of extreme temperature (hot and cold) and humidity

(wet and dry), and (3) elucidate the variations in yield sensitivity to compound events of extreme temperature (hot and cold) and humidity (wet and dry) across different wheat varieties and sowing dates.

## 2. Methods

### 2.1. Nationwide wheat yield traits observations

The China Meteorological Center (CMC) managed the winter wheat yield traits data obtained from the nationwide agrometeorological observation network. A total of 85 stations, strategically located in the primary wheat-producing regions throughout the country, were selected for analysis (Fig. 1). These stations were chosen due to their significance, collectively representing over 50 % of the total wheat cultivation area and being prone to extreme weather conditions. Specifically, Gansu (GS), Shaanxi (SAX), and Shanxi (SX) fall within warm temperate, semi-humid, and semi-arid climate zones, characterized by typical features of a continental monsoon climate. Throughout the winter wheat growing period, GS recorded an average temperature, relative humidity (RH), and cumulative precipitation of 7.4 °C, 59.8 %, and 192.2 mm, respectively. In SAX, these values were 8.0 °C, 63.7 %, and 250.3 mm, while SX experienced 9.5 °C, 72.3 %, and 326.3 mm (Table S1). Conversely, Henan (HN), Hebei (HB), and Shandong (SD) primarily encounter a temperate monsoon climate in semi-humid areas. During the winter wheat growing season, the average RH, temperature, and cumulative precipitation were 9.5 °C, 71.0 %, and 315.6 mm in HN, 8.5 °C, 58.1 %, and 184.9 mm in HB, and 6.4 °C, 53.8 %, and 166.1 mm in SD (Table S1). The majority of these observation sites have recorded data spanning 13 to 30 years, encompassing various yield-related traits such as canopy height, final ear density, grain number per ear, grain weight, and grain filling rate (Eq. 1). It is worth noting that these traits were documented in most stations located in HN, HB, and SD, but not consistently in GS, SAX, and SX (Table S1).

$$GFR = \frac{GW}{GDD_{GF-MT}} \quad (1)$$

where  $GFR$ ,  $GW$ , and  $GDD_{GF-MT}$  are the grain filling rate ( $g\ ^\circ C^{-1}\ day^{-1}$ ), grain weight (g), and growing degree day during grain filling and maturity ( $^\circ C\ day$ ), respectively.

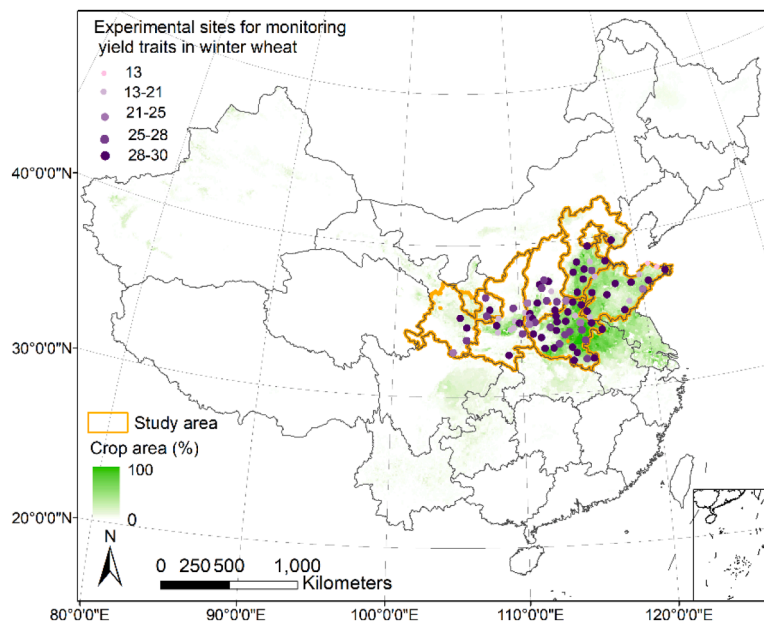


Fig. 1. Spatial distribution of 85 wheat yield trait observation sites and their time span. The purple dots represent the distribution of wheat trait observation stations, with darker colors indicating longer planting durations. Green grids depict the proportion of wheat planting, measured in percentage. Yellow polygons denote the primary regions of station distribution.

**Table 1**

Abbreviations of climate indices and wheat yield traits.

| Abbreviation  | Full name                                               | Unit                                 |
|---------------|---------------------------------------------------------|--------------------------------------|
| HDD           | Hot-dry days                                            | day                                  |
| CWD           | Cold-wet days                                           | day                                  |
| RH            | Relative humidity                                       | %                                    |
| $T_{min}$     | Minimum temperature                                     | °C                                   |
| $T_{max}$     | Maximum temperature                                     | °C                                   |
| Prdp          | Precipitation                                           | mm                                   |
| GDD           | Growing degree day                                      | °C day                               |
| EDD           | Extreme growing degree days                             | °C day                               |
| FDD           | Freezing degree-day                                     | °C day                               |
| $T_h$         | The temperature corresponding to hour h                 | °C                                   |
| $T_{base}$    | Specific base temperature                               | °C                                   |
| $T_{opt}$     | Optimum temperature                                     | °C                                   |
| $T_{frez}$    | Freeze injury temperature                               | °C                                   |
| H             | Hour                                                    | h                                    |
| PT-JT         | Planting to jointing                                    | —                                    |
| JT-HD         | Jointing to heading                                     | —                                    |
| HD-MT         | Heading to maturity                                     | —                                    |
| HD-FL         | Heading to flowering                                    | —                                    |
| GF-MT         | Grain filling to maturity                               | —                                    |
| TW            | Time window                                             | kg ha <sup>-1</sup>                  |
| Y             | Yield                                                   | —                                    |
| YT            | Yield trait                                             | —                                    |
| GFR           | grain filling rate                                      | g °C <sup>-1</sup> day <sup>-1</sup> |
| GW            | grain weight                                            | g                                    |
| CI            | Climate indices                                         | —                                    |
| NC            | Number of cultivars                                     | —                                    |
| $CI_j^r$      | Standard climate indices j                              | —                                    |
| $CI_j$        | climate indices j                                       | —                                    |
| $CI_j^{mean}$ | Mean of climate indices j                               | —                                    |
| $CI_j^{sd}$   | Standard deviation of the corresponding climate indices | —                                    |

## 2.2. Climate indices

We obtained daily climate data, including relative humidity (RH, %), minimum temperature ( $T_{min}$ , °C), maximum temperature ( $T_{max}$ , °C), and precipitation (Prdp, mm), from the experimental sites operated by CMC in this study. The cosine curve (Eq. 2) was used to fit the hourly temperature based on the daily  $T_{min}$  and  $T_{max}$ . Subsequently, the growing degree day (GDD, °C day, Eq. 3), the extreme growing degree days (EDD, °C day, Eq. 4), and the freezing degree-day (FDD, °C day, Eq. 5) were calculated for each station.

$$T_h = \left( \frac{T_{max} + T_{min}}{2} \right) + \left( \frac{(T_{max} - T_{min}) \times \cos(0.216 \times (h - 14))}{2} \right) \quad (2)$$

$$GDD = \sum_{d=1}^{d=n} \sum_{h=1}^{h=24} GDD_{dh}^d; GDD_h = \begin{cases} 0 & T_h < T_{base} \\ \frac{(T_h - T_{base})}{24} & T_{base} \leq T_h \leq T_{opt} \\ \frac{(T_{opt} - T_{base})}{24} & T_h > T_{opt} \end{cases} \quad (3)$$

$$EDD = \sum_{d=1}^{d=n} \sum_{h=1}^{h=24} EDD_{dh}^d; EDD_h = \begin{cases} 0 & T_h \leq T_{opt} \\ \frac{T_h - T_{opt}}{24} & T_h > T_{opt} \end{cases} \quad (4)$$

$$Y = \beta_1 EDD_{HD-MT} + \beta_2 FDD_{PT-JT} + \beta_3 Prdp_{PT-JT} + \beta_4 Prdp_{PT-JT}^2 + \beta_5 Prdp_{JT-HD} + \beta_6 Prdp_{JT-HD}^2 + \beta_7 Prdp_{HD-MT} + \beta_8 Prdp_{HD-MT}^2 + \beta_9 CWD_{JT-HD} + \beta_{10} CWD_{HD-MT} + c_{CWD} + y_{CWD} + \varepsilon_{CWD} \quad (6)$$

$$Y = \beta_1 EDD_{HD-MT} + \beta_2 FDD_{PT-JT} + \beta_3 Prdp_{PT-JT} + \beta_4 Prdp_{PT-JT}^2 + \beta_5 Prdp_{JT-HD} + \beta_6 Prdp_{JT-HD}^2 + \beta_7 Prdp_{HD-MT} + \beta_8 Prdp_{HD-MT}^2 + \beta_9 HDD_{HD-MT} + c_{HDD} + y_{HDD} + \varepsilon_{HDD} \quad (7)$$

$$FDD = \sum_{d=1}^{d=n} \sum_{h=1}^{h=24} FDD_{dh}^d; FDD_h = \begin{cases} \frac{T_h - T_{frez}}{24} & T_h \leq T_{frez} \\ 0 & T_h > T_{frez} \end{cases} \quad (5)$$

where  $T_{frez}$ ,  $T_{base}$ , and  $T_{opt}$  represent the freeze injury, specific base, and optimum temperature thresholds associated with each growth stage of wheat, as detailed in Table S2 (Zhang et al., 2022). More full names of the abbreviation see Table 1.

Generally, the optimal lower limit of temperature for winter wheat ranges around 5 °C during the jointing to grain-filling stage (Luo, 2011; Zhang et al., 2022). The incidence of diseases in winter wheat tends to increase when humidity levels exceed 60 % (El Jarroudi et al., 2017; Junk et al., 2016). Thus, the cold-wet days (CWD) were defined as days with  $T_{min} < 5$  °C,  $RH > 85$  %, and precipitation  $> 0.1$  mm in this study. Similarly, the hot-dry days (HDD) were defined as days within a specific time window where the  $T_{max} > 30$  °C,  $RH < 35$  %, and both  $T_{max}$  and  $RH$  showed a trend of increase and decrease (Liu et al., 2014; Luo, 2011; Porter and Gawith, 1999; Zhao et al., 2022).

## 2.3. Trend analysis

In our study, we analyzed the trends of climate indices (EDD, FDD, CWD, and HDD) from 1981 to 2010 using ordinary least-square regression. Statistical significance was consistently evaluated through Mann-Kendall analysis at a 95 % confidence level. However, due to the count nature and Poisson distribution (Figure S1) of CWD and HDD, we used Poisson regression methods to examine the trends in these variables (Zhao et al., 2022).

## 2.4. Climate indices-driven yield trends

Then, we quantified climate indices-driven yield trends by combining climate indices trends with their corresponding yield sensitivities. These yield sensitivities, or regression coefficients, were estimated using linear mixed regression models based on the original climate indices. The 95 % confidence intervals were determined by the bootstrapping method, which involves generating a set of models by resampling (replacing) the observed data 1000 times. The 2.5th and 97.5th percentiles were selected from the 1000 bootstrap replications as the 95 % confidence intervals for each regression coefficient. In these models, a panel dataset with repeated measures of X and Y over time was employed to account for unobserved heterogeneity that remains constant over time. Fixed effects of wheat varieties were introduced in place of stations to capture this unobserved heterogeneity (Lobell et al., 2020). Moreover, to eliminate the impact of covariance, sensitivity analysis for yield traits was independently conducted for both CWD (Eq. 6) and HDD (Eq. 7). The climate index-driven yield trends were then obtained by multiplying the regression coefficients of the respective climate indices with their time trends (Eq. 8). To enhance the robustness of the model, we defined CWD ( $T_{min}$ : 4 °C, 5 °C, and 6 °C; RH: 80 %, 85 %, 90 %) and HDD ( $T_{max}$ : 30 °C, 31 °C, 32 °C; RH: 30 %, 35 %, 40 %) using a combination of nine-group threshold values (Figure S3).

$$Y_{Trend,ci} = \beta_{ci} * Trend_{ci} * 10 \quad (8)$$

where  $\beta_1 \dots \beta_{10}$  are the regression coefficients of corresponding original unit climate indices;  $C_{CWD}$  and  $C_{HDD}$  are cultivar-specific vectors in original CWD and HDD models;  $y_{CWD}$  and  $y_{HDD}$  are time-random effect vector in original CWD and HDD models;  $y_{CWD}$ , and  $\varepsilon_{HDD}$  are the random error term in original unit CWD and HDD model;  $PT-JT$ ,  $JT-HD$ , and  $HD-MT$  are time window of planting to jointing, jointing to heading, and heading to maturity, respectively.

## 2.5. Yield traits anomalies of CWD and HDD

For a more detailed exploration of how anomalies in yield traits (YT) respond to CWD<sub>JT-MT</sub> (form jointing to maturity) and HDD<sub>HD-MT</sub> (from heading to maturity), we computed the changes in YT associated with station  $i$ , traits  $j$ , cultivar  $k$ , and climate indices  $CI$  using the station-cultivar-year search method (Eq. 9) (Fu et al., 2023; Noia Junior et al., 2023). We identified unusual yield traits by comparing instances where CWD and HDD occurred with specific varieties at site  $i$  (cultivated for at least 5 years) to instances where CWD and HDD did not occur with the same varieties at the same site.

$$YT_{(CI,i,j)} = \frac{1}{NC_{(i,j)}} \sum_{k=1}^{k=n} \frac{YT_{TW(i,j,k)} - YT_{CI,TW(i,j,k)}}{YT_{TW(i,j,k)}} \quad (9)$$

where  $YT_{(CI,i,j)}$  is the change of yield traits  $j$  when climate indices  $CI$  occurred at  $i$ th research station;  $NC$  is the number of cultivars;  $k$  is the number of seasons corresponding to planted cultivars;  $YT_{TW(i,j,k)}$  and  $YT_{CI,TW(i,j,k)}$  are the yield traits  $j$  of the corresponding varieties when CWD and HDD did and did not occur at site  $i$ , respectively.

## 2.6. Yield traits sensitivities to standard climate indices

To account for variations in different yield traits (such as canopy height, final ear density, grain number per ear, grain weight, grain filling rate, and grain yields) sensitivities, we utilized standardized climate indices (EDD, FDD, CWD, and HDD) in a linear mixed regression model (Eq. 10-12). The regression coefficients obtained from these standardized models clarify the specific impacts of climate indices on yield traits. This method has been used to quantify the impacts of extreme climate on wheat and maize yields (Li et al., 2019; Zhao et al., 2022). Simply, climate indices  $j$  were standardized ( $CI_j^\sigma$ ) as the difference between the corresponding climate indices ( $CI_j$ ) and its multi-year average ( $CI_j^{mean}$ )

divided by one standard deviation ( $CI_j^{sd}$ ) of the corresponding climate indices (Eq. 10). Given the varied temporal spans of different yield-related traits, various yield trait models utilize different scaled climate indices (Table 2). Moreover, threshold values for nine groups, aligned with CWD and HDD, were implemented in the models for yield traits. As with the original unit model, the bootstrapping method was used to determine 95 % confidence intervals for the standard unit models.

$$CI_j^\sigma = \frac{CI_j - CI_j^{mean}}{CI_j^{sd}} \quad (10)$$

$$YT_j = \beta_j^\sigma X_j^\sigma + c_j^\sigma + y_j^\sigma + \varepsilon_j^\sigma \quad (11)$$

$$YT_{ij}^s = \frac{\beta_{ij}^\sigma}{YT_j} \times 100\% \quad (12)$$

where  $YT_j$  is the  $j$ th yield traits (Table 2);  $\beta_j^\sigma$  is the regression coefficients of  $j$ th yield traits to standard climate indices (Table 2);  $c_j^\sigma$ ,  $y_j^\sigma$ , and  $\varepsilon_j^\sigma$  are cultivar-specific vector, time-fixed effect vector, and random error, respectively;  $YT_{ij}^s$  is the  $j$ th yield traits sensitivities to  $i$ th climate indices, %.

## 2.7. Yield sensitivity to standard climate indices of different varieties and sowing date

To explore the impact of varieties and agronomic practices on the sensitivity of yield to standard climate indices, we performed regression analyses (Eq. 10-12) using yield clumps from various varieties or agronomic practices and standard climate indices. For varieties, we examined datasets containing early (171 ~ 280 day after sowing), medium (194 ~ 294 day after sowing), and late maturing (210 ~ 308 day after sowing) varieties for regression. Given the influence of the sowing period on regulating the heat time during crop growing days (Minoli et al., 2022), the agronomic measures primarily focused on the sowing period. Accordingly, we screened data for regression based on early sowing (251 ~ 306 day of year), normal or medium sowing (260 ~ 323 day of year), and late sowing (267 ~ 341 day of year). Sowing and ripeness classification relied on equal differences in the long-term recorded sowing date and growing days of respective stations.

## 3. Results

### 3.1. Extreme climate indices exposure and temporal trends

Firstly, we investigated the exposure and temporal trends of various extreme climate indices (CWD<sub>JT-HD</sub>, CWD<sub>HD-MT</sub>, and HDD<sub>HD-MT</sub> in Fig. 2, EDD<sub>HD-MT</sub>, FDD<sub>PT-JT</sub> in Figure S2) for winter wheat in the China wheat belt. The average CWD<sub>JT-HD</sub> ranged from 0 to 4 days, with a higher exposure potential in SAX and HN (Fig. 2a). However, the trends exhibited a decline (-1.6 ~ -0.5 days dec<sup>-1</sup>) in SAX and HN, and a slight increase (0.25 days dec<sup>-1</sup>) in SD and SX (Fig. 2b). The mean and trend of CWD from heading to maturity were smaller than that from jointing to heading (Fig. 2c-d). Both CWD<sub>JT-HD</sub> and CWD<sub>HD-MT</sub> showed no significant trends across most stations (Fig. 1b, 2d). In contrast, the spatial pattern of HDD<sub>HD-MT</sub> displayed an inverse distribution to CWD<sub>JT-HD</sub> and CWD<sub>HD-MT</sub>, where HDD<sub>HD-MT</sub> was prominent in areas with smaller CWD<sub>JT-HD</sub> and CWD<sub>HD-MT</sub> (Fig. 2e). The average number of HDD<sub>HD-MT</sub> ranged from 0–1.5 days (Fig. 2e). Spatial trends of HDD<sub>HD-MT</sub> indicated a westward increase and eastward decrease, or a trend of increasing HDD<sub>HD-MT</sub> where the HDD value is large (Fig. 2f).

The multi-year averages of EDD<sub>HD-MT</sub> ranged from 5 ~ 30 °C day, with a change rate of 0 to 20 °C day decade<sup>-1</sup> (Figure S2a-b) across the 85 stations. Notably, GS, SAX, SX, and HN stations showed more significant changes compared to HB and SD. The averaged FDD<sub>PT-JT</sub> displayed a

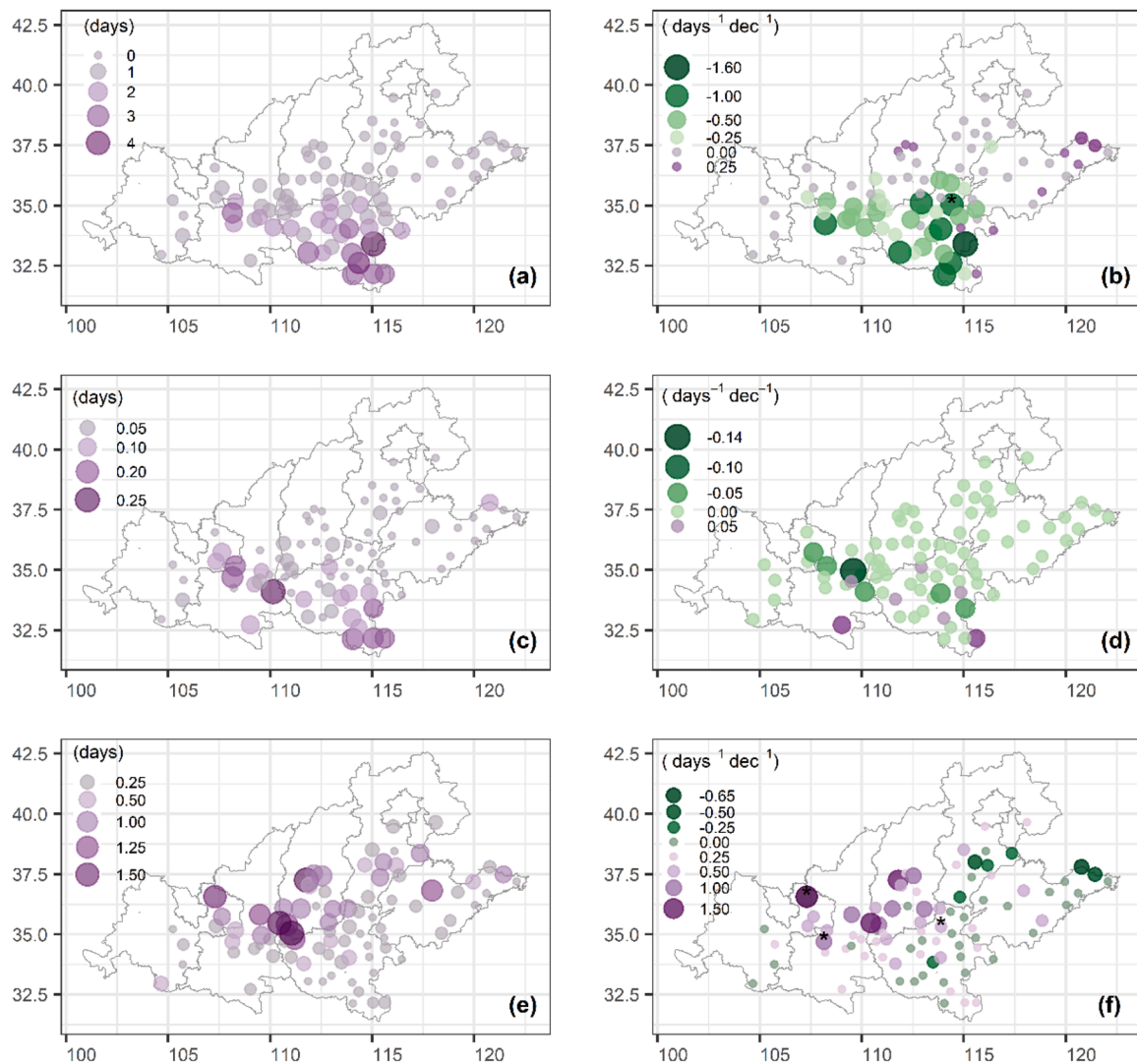
**Table 2**  
Standard climate indices corresponding to yield traits in the models.

| Traits                          | Climate indices                                                                                                                                                   |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Grain yield <sup>a</sup>        | $EDD_{HD-MT}^\sigma, FDD_{PT-JT}^\sigma, \sum_{TW=1}^{TW=3} (Prp_{TW}^\sigma + (Prp_{TW}^\sigma)^2),$<br>$\sum_{TW=2}^{TW=3} CWD_{TW}^\sigma, HDD_{HD-MT}^\sigma$ |
| Grain weight <sup>a</sup>       | $EDD_{HD-MT}^\sigma, FDD_{PT-JT}^\sigma, \sum_{TW=1}^{TW=3} (Prp_{TW}^\sigma + (Prp_{TW}^\sigma)^2),$<br>$\sum_{TW=2}^{TW=3} CWD_{TW}^\sigma, HDD_{HD-MT}^\sigma$ |
| Grain number <sup>b</sup>       | $EDD_{HD-FL}^\sigma, FDD_{PT-JT}^\sigma, \sum_{TW=1}^{TW=3} (Prp_{TW}^\sigma + (Prp_{TW}^\sigma)^2),$<br>$\sum_{TW=2}^{TW=3} CWD_{TW}^\sigma, HDD_{HD-FL}^\sigma$ |
| Ear density <sup>b</sup>        | $EDD_{HD-FL}^\sigma, FDD_{PT-JT}^\sigma, \sum_{TW=1}^{TW=3} (Prp_{TW}^\sigma + (Prp_{TW}^\sigma)^2),$<br>$\sum_{TW=2}^{TW=3} CWD_{TW}^\sigma, HDD_{HD-FL}^\sigma$ |
| Grain filling rate <sup>a</sup> | $EDD_{HD-MT}^\sigma, FDD_{PT-JT}^\sigma, \sum_{TW=1}^{TW=3} (Prp_{TW}^\sigma + (Prp_{TW}^\sigma)^2),$<br>$\sum_{TW=2}^{TW=3} CWD_{TW}^\sigma, HDD_{HD-MT}^\sigma$ |
| Canopy height <sup>a</sup>      | $FDD_{PT-JT}^\sigma, \sum_{TW=1}^{TW=3} (Prp_{TW}^\sigma + (Prp_{TW}^\sigma)^2), CWD_{JT-HD}^\sigma$                                                              |

<sup>a</sup> TW (time widow) include of  $PT-JT$ ,  $JT-HD$ , and  $HD-MT$  corresponding to 1, 2, and 3 in yield, Grain weight, and Canopy height models.

<sup>b</sup> TW of  $PT-JT$ ,  $JT-HD$ , and  $HD-FL$  corresponding to 1, 2, and 3 in Grain number and Ear density models.





**Fig. 2.** Averages and trends of CWD<sub>JT-HD</sub> (a, b), CWD<sub>HD-MT</sub> (c-d), and HDD<sub>HD-MT</sub> days (e-f) of winter wheat from 1981–2010 in China wheat belt. “\*\*” in Fig. 2 (b, d, and f) indicated the significant ( $P < 0.001$ ) change.

distinct spatial pattern, with higher values in the north (Figure S2c: 20 ~ 60 °C day) than in the south (Figure S2c: 0 ~ 20 °C day). The trend of FDD<sub>PT-JT</sub> contrasted with that of EDD<sub>HD-MT</sub>, with the majority of sites displaying a downward trend, especially in the northern part of the country (Figure S2d). Overall, the likelihood of extreme compound events, involving cold-wet and hot-dry, was higher in GS, SAX, SX, and HN compared to HB and SD.

### 3.2. Yield traits anomalies of CWD and HDD

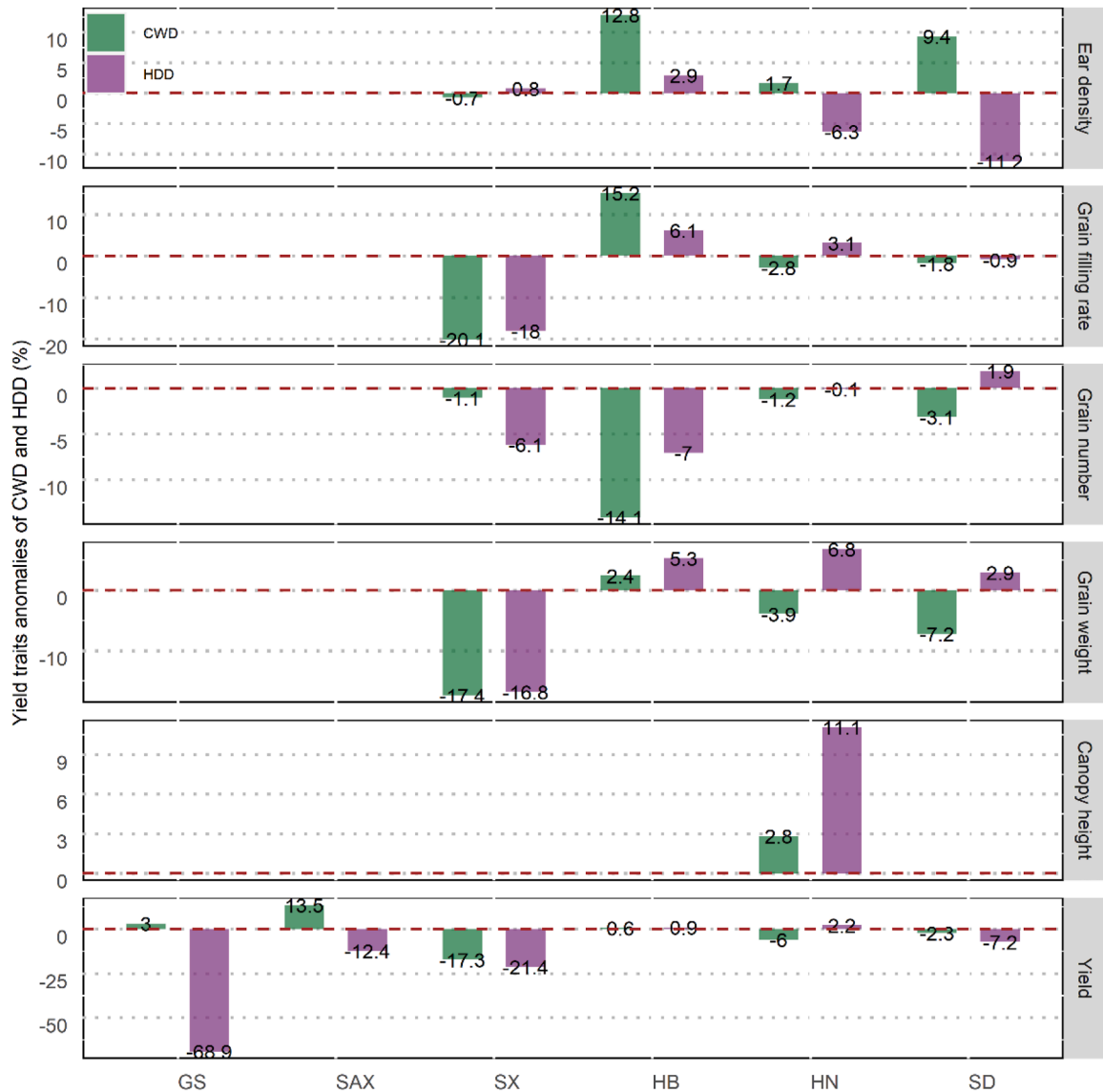
Considering the significant exposure of wheat to the combined conditions of cold-wet and hot-dry weather, we evaluated anomalies in yield traits under occurrences of CWD and HDD (Fig. 3). Except for GS and SAX, wheat yield experienced a decline due to CWD, with values of -17.3 %, -6.0 %, and -2.3 % in SX, HN, and SD, respectively (Fig. 3). Areas, where crop yield remained unaffected by CWD conditions, were more susceptible to yield shocks caused by HDD events. Yield reductions attributed to HDD were documented as -68.9 %, -12.4 %, -21.4 %, and -7.2 % in GS, SAX, SX, and SD, respectively (Fig. 3).

Analyzing specific yield traits, grain number was adversely affected by CWD in most regions, showing reductions of -1.1 % in SX, -14.1 % in HB, -1.2 % in HN, and -3.1 % in SD. Ear density was more susceptible to losses from HDD (-11.2 ~ 2.9 %) than CWD (-0.7 ~ 12.8 %). The impacts

of CWD and HDD on grain weight and grain filling rate varied among regions. In summary, wheat yield across the China wheat belt encountered disruptions caused by Cold-Wet (CWD) and Hot-Dry (HDD) events from 1981 to 2010. The fluctuations in yield were primarily attributed to the varied responses of different traits to CWD and HDD in different regions.

### 3.3. Extreme climate indices-driven yield changes

Subsequently, we quantified climate indices-driven trends in yield by combining the trends in climate indices (from trend analysis) with the sensitivity of yield traits to these original climate indices (Table S3). The models could explain 65 % of the variance in yield and demonstrate that almost all combinations of temperature and humidity thresholds for CWD and HDD caused yield loss (Figure S3: -697.42 ~ 21.45 kg ha<sup>-1</sup> day<sup>-1</sup>). Despite the negative sensitivity of yield response to FDD<sub>PT-JT</sub> and CWD<sub>JT-HD</sub> (Table S3), there were no increases in the yield trends driven by FDD<sub>PT-JT</sub> and CWD<sub>JT-HD</sub> (Fig. 3a) due to the declining trends (Figure S2d and Fig. 2b). The EDD<sub>HD-MT</sub> exhibited a declining trend at almost all sites, with a mean rate of -9.80 kg ha<sup>-1</sup> decade<sup>-1</sup> (Fig. 4a). Spatially, EDD<sub>HD-MT</sub> caused more pronounced yield decreases in GS, SAX, SX, and HN, with reductions ranging from 11.14 ~ 114.94 kg ha<sup>-1</sup> decade<sup>-1</sup>. (Fig. 4b). On average, the decline in yield driven by



**Fig. 3.** Yield traits (ear density, grain filling rate, grain number, grain weight, canopy height, and yield) anomalies of CWD and HDD at stations in GS, SAX, SX, HB, HN, and SD, respectively.

HDD<sub>HD-MT</sub> (Fig. 4a:  $-23.24 \text{ kg ha}^{-1} \text{ decade}^{-1}$ ) was larger than that driven by EDD<sub>HD-MT</sub> (Fig. 4a:  $-9.80 \text{ kg ha}^{-1} \text{ decade}^{-1}$ ). In addition, the magnitude of the yield trend driven by HDD<sub>HD-MT</sub> was larger than that of the driven by EDD<sub>HD-MT</sub> in GS, SAX, SX, and HN (Fig. 4c:  $-280.96 \sim -53.40 \text{ kg ha}^{-1} \text{ decade}^{-1}$ ). Considering this, trends in yield due to climate indices varied based on their trends and the sensitivity of yield to them.

### 3.4. Yield traits sensitivity to the standard unit of extreme climate indices

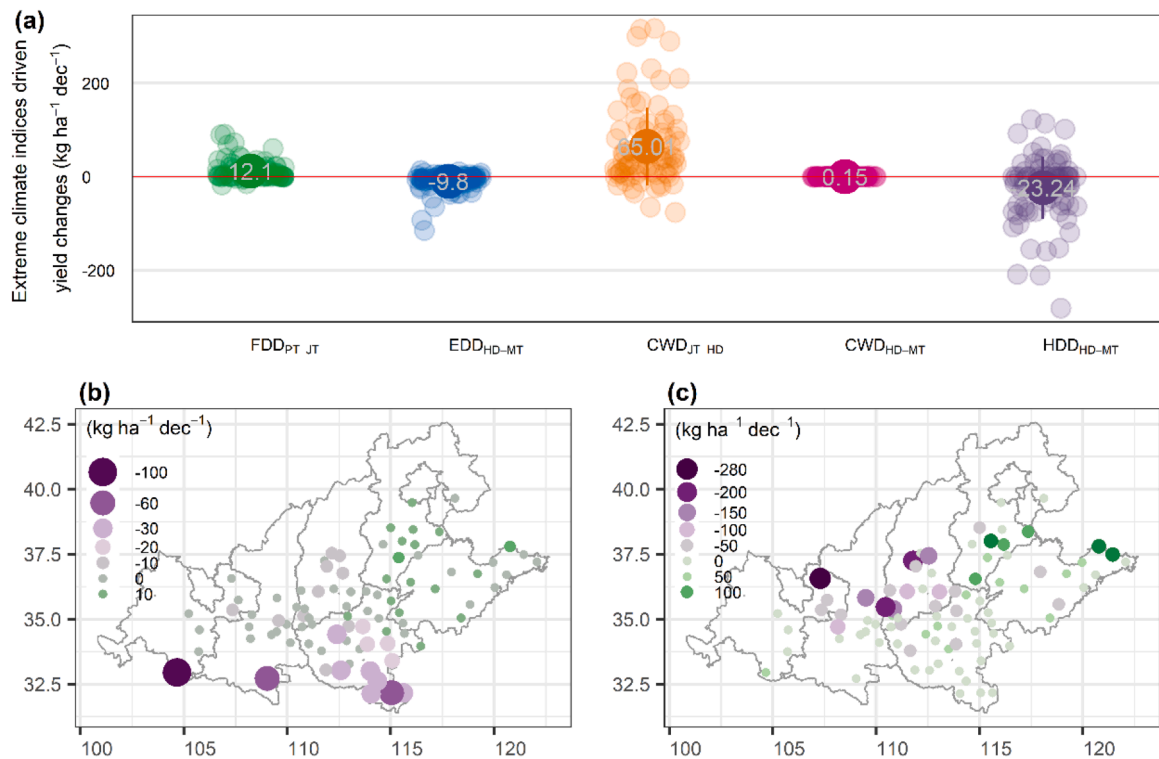
Following that, we employed standardized regression modeling, incorporating threshold values for nine groups associated with CWD and HDD, to assess the sensitivities of yield traits to different extreme climate indices. These models demonstrated their stability across different thresholds (Figure S4–9), revealing variations of 65 %, 61 %, 65 %, 61 %, 59 %, and 64 % of the variability in yield, grain weight, grain number, ear density, grain filling rate, and canopy height, respectively (Table S4–9). The highest sensitivity to yield loss was observed in HDD<sub>HD-MT</sub>, showing a decrease of 1.90 % per standard unit (Fig. 5a). CWD<sub>HD-MT</sub> demonstrated yield sensitivity closely following HDD<sub>HD-MT</sub>, with a value of  $-1.75 \%$ . Moreover, our models indicated almost similar yield loss between EDD<sub>HD-MT</sub> and CWD<sub>JT-HD</sub> (Fig. 5a). Notability, there

was a larger yield sensitivity to FDD<sub>PT-JT</sub> than that to EDD<sub>HD-MT</sub>.

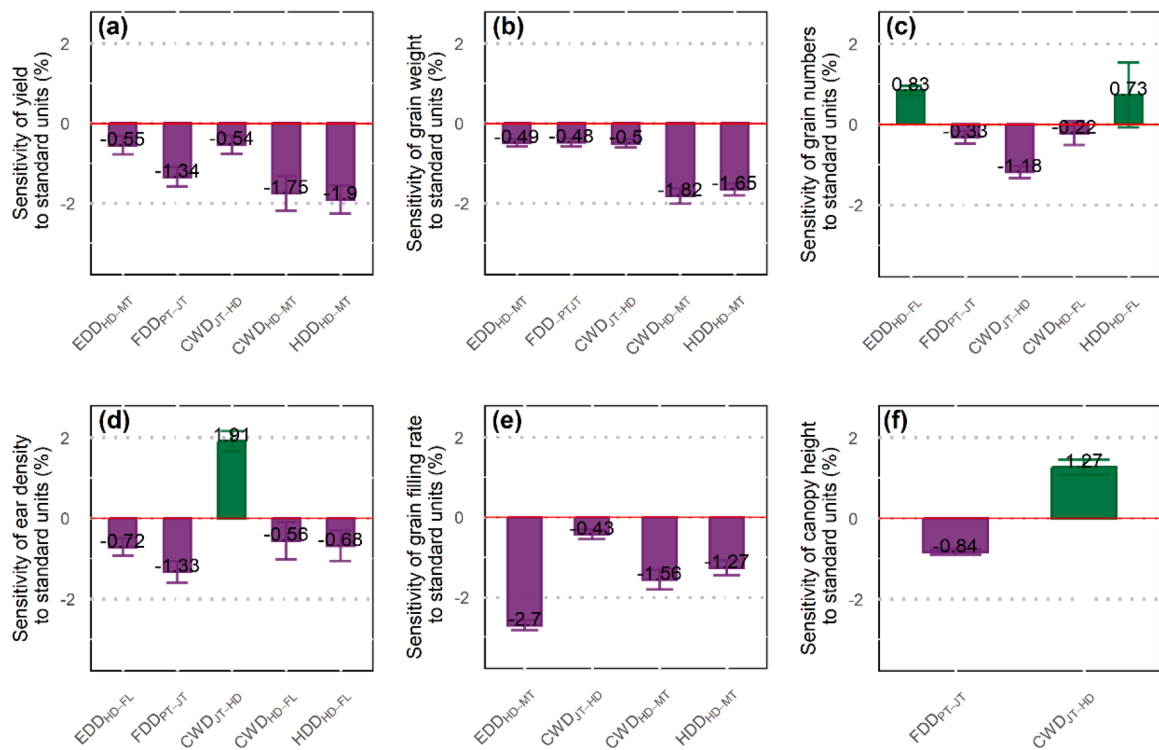
Among all yield traits, grain weight was the only one that could be affected by all extreme climate indices (Fig. 5b). CWD<sub>HD-MT</sub> attributed the largest decline of grain weight in the standard regression models, with a value of  $-1.82 \%$  (Fig. 5b). Grain number loss showed more sensitivity to cold (Fig. 5c:  $-0.33 \%$ ) or cold-wet compound (Fig. 5c:  $-1.18 \sim -0.22 \%$ ) than hot (Fig. 5c:  $0.83 \%$ ) or hot-dry compound (Fig. 5c:  $0.73 \%$ ). Ear density decline was more sensitive to FDD<sub>PT-JT</sub> (Fig. 5d:  $-1.33 \%$ ) than EDD<sub>HD-FL</sub> (Fig. 5d:  $-0.72 \%$ ), HDD<sub>HD-FL</sub> (Fig. 5d:  $-0.68 \%$ ), and CWD (Fig. 5d:  $-0.56 \sim 1.91 \%$ ). The averaged grain filling rate changes were more sensitive to EDD<sub>HD-MT</sub> (Fig. 5e:  $-2.70 \%$ ) than to CWD<sub>JT-HD</sub> (Fig. 5e:  $-0.43 \%$ ), CWD<sub>HD-MT</sub> (Fig. 5e:  $-1.56 \%$ ), and HDD<sub>HD-MT</sub> (Fig. 5e:  $-1.27 \%$ ). Canopy height showed a greater reduction by the extreme climate indices during the early stage of winter wheat (Fig. 5f).

### 3.5. Yield sensitivity to extreme indices under different varieties and sowing date

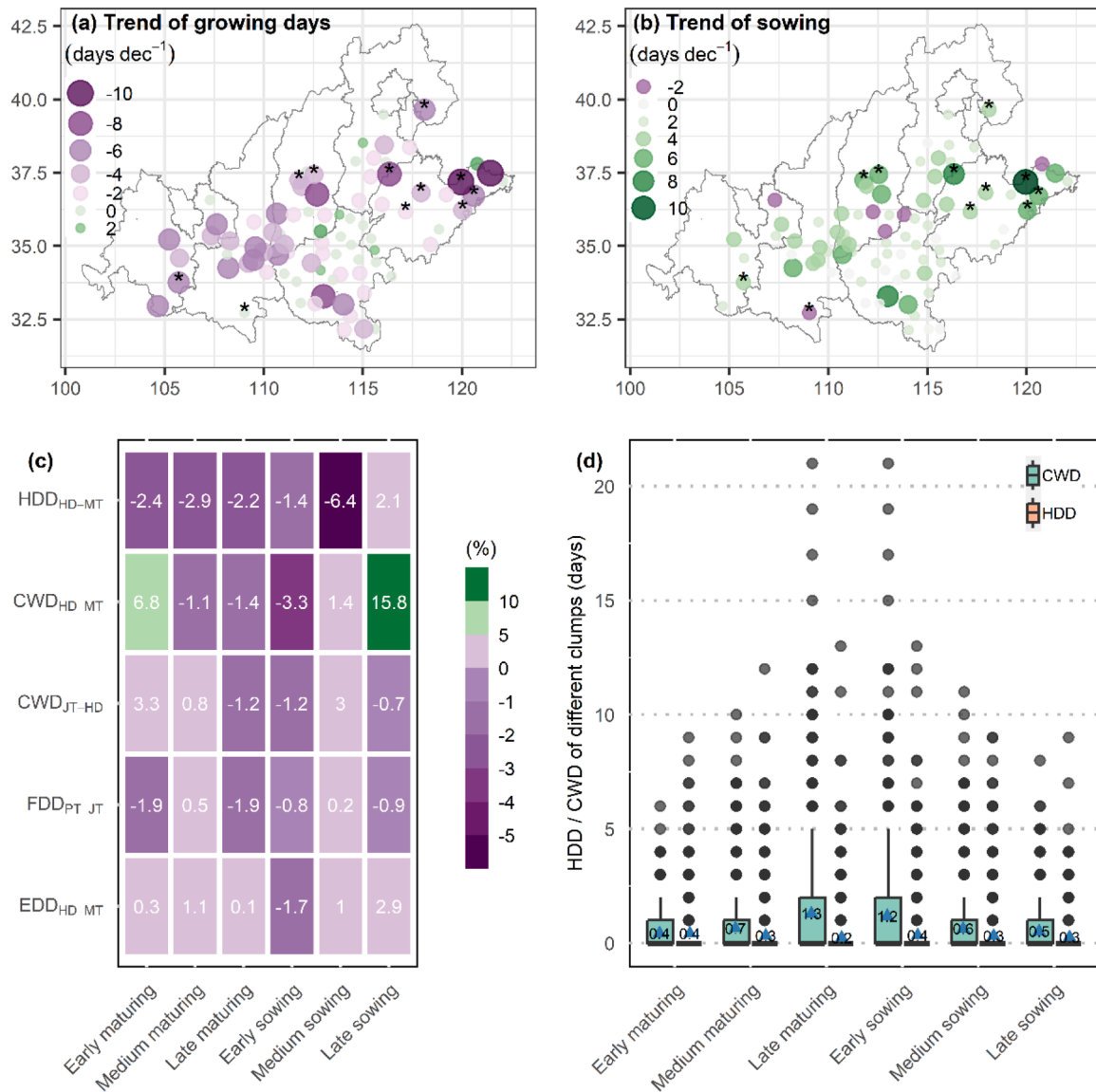
Lastly, utilizing models for different varieties and sowing clumps, we investigated the differences in wheat yield sensitivity to extreme indices



**Fig. 4.** Rates of yield trends driven by extreme climate indices of winter wheat from 1981–2010 in China wheat belt (a) and the spatial pattern of yield change rates driven by EDD<sub>HD-MT</sub> (b) and HDD<sub>HD-MT</sub> (c).



**Fig. 5.** Yield traits (a: grain yield; b: grain weight; c: grain number; d: ear density; e: grain filling rate; f: canopy height) sensitivities respond to the standard unit of climate indices (EDD<sub>HD-FL</sub>, EDD<sub>HD-MT</sub>, FDD<sub>PT-JT</sub>, CWD<sub>JT-HD</sub>, CWD<sub>HD-FL</sub>, CWD<sub>HD-MT</sub>, HDD<sub>HD-FL</sub>, HDD<sub>HD-MT</sub>). One standard unit of EDD<sub>HD-FL</sub>, EDD<sub>HD-MT</sub>, FDD<sub>PT-JT</sub>, CWD<sub>JT-HD</sub>, CWD<sub>HD-FL</sub>, CWD<sub>HD-MT</sub>, HDD<sub>HD-FL</sub>, and HDD<sub>HD-MT</sub> across the study domain is 9.05 °C day, 9.65 °C day, 15.18 °C day, 1.67 days, 0.18 days, 0.19 days, 0.24 days, and 0.68 days, respectively.



**Fig. 6.** Trends in growing days (a) and sowing date (b) in 1981–2010, the sensitivity of yields to extreme climate indices under different varieties and sowing date clumps (c), and the number of days of CWD and HDD under different varieties and sowing date clumps in China winter wheat belt.

under different varieties and sowing dates. From 1981 to 2010, wheat growing days were shortened by 0–10 days and sowing was delayed by 0–10 days at most stations in China wheat belt (Fig. 6a–b). Concerning varieties, yield changes caused by CWD<sub>JT-HD</sub> (–1.2 ~ 3.3 %) and CWD<sub>HD-MT</sub> (–1.4 ~ 6.8 %) could increase with the extension of wheat growing days (Fig. 6c). However, the extension of wheat growing days was less effective in responding to HDD<sub>HD-MT</sub>, with yield changes of –2.4 %, –2.9 %, and –2.2 % for early-maturing, medium-maturing, and late-maturing varieties, respectively (Fig. 6c). Additionally, there were fewer HDD<sub>HD-MT</sub> and CWD in the model for early-maturing varieties compared to medium-maturing and late-maturing varieties (Fig. 6d).

Regarding the sowing date, yield changes caused by CWD<sub>JT-HD</sub> (–1.2 ~ 3.0 %) and CWD<sub>HD-MT</sub> (–3.3 ~ 15.8 %) could increase with the postponement of the wheat sowing date (Fig. 6c). Moreover, delaying the wheat sowing date help mitigate yield losses caused by HDD<sub>HD-MT</sub> (Fig. 6c). Furthermore, there were fewer CWD instances in the late-sowing model compared to early-sowing and medium-sowing clumps (Fig. 6c). In summary, the model proposed that the impact of extreme indices on yield loss in early-maturity wheat was less than that in the medium and late-maturity wheat. Likewise, it indicated that the impacts of extreme indices on yield loss in late-sown wheat was lower than in

early and medium sowing wheat.

#### 4. Discussion

Generally speaking, the interaction between crops and the atmosphere is a crucial pathway to understand the comprehensive effects of extreme heat and humidity on global crop yields under climate change (Lesk et al., 2021). Based on long-term monitoring network for winter wheat traits in the field, we employed linear mixed regression models to quantify the sensitivity of wheat yield traits to compound cold-wet and hot-dry atmospheric conditions in the China wheat belt. Our analysis not only quantified the yield losses from CWD and HDD (Fig. 5a) but also unveiled variations in the sensitivity of different yield traits to CWD and HDD (Fig. 5b–f). Our findings indicated that wheat grain weight was adversely affected by all extreme climate indices, especially when HDD occurred from the heading to the maturity stage (Fig. 5b). Conversely, wheat grain number demonstrated higher sensitivity to CWD during the period between jointing and maturity (Fig. 5c). Unfortunately, recent crop-climate-model-based climate change studies have largely overlooked the compounding effects of atmospheric temperature extremes and humidity extremes on crop yields (Jägermeyr et al., 2021; Jiang



et al., 2022; Liu et al., 2023). While some studies have explored the effects of hot and dry atmospheric extremes on crop yields using linear mixed regression models (Hamed et al., 2021; Zampieri et al., 2017; Zhao et al., 2022). For example, Zhao et al. (2022) calculated the yield losses of U.S. winter wheat associated with compound hot-dry-windy events, emphasizing the predominant attention on the consequences of extreme heat and drought. Importantly, the influence of the simultaneous occurrence of extreme low temperatures and extreme humidity in the atmosphere on yields has been overlooked. Earlier studies have underscored that extremes in atmospheric humidity and low temperatures can result in crop seed failures and diseases, ultimately resulting in diminished crop yields (Ben-Ari et al., 2018; N6ia J6nior et al., 2023).

Wheat yield in HN was not damaged by the presence of HDD, which appears contradictory to findings from previous studies (Perdomo et al., 2015; Prasad et al., 2011). This could be attributed to lower exposure to HDD (Fig. 2e: 0~0.5 day) during the wheat growing season in HN. Our study emphasized that the greater the exposure to HDD, the larger the yield loss in the corresponding region. (Fig. 2-3). Moreover, Lesk et al. (2021) also demonstrated that maize and soy yields dropped more in areas with reduced rainfall, greater evapotranspiration, and higher temperatures. In addition, our study emphasized the yield loss of winter wheat due to extreme wet and cold in most region. However, it was found that wheat yield increased with occurrences of CWD in GS and SAX. The lower CWD exposure during the winter wheat growing season in GS and SAX, combined with the dry climate in GS and SAX (Tao et al., 2014), might lead to less negative impacts of CWD on yields, and even yield increases.

Standard linear mixed regression models demonstrated that wheat yield changed by -2.25 % per standard (1 standard deviation) of  $HDD_{HD-MT}$  (Fig. 5a), aligning with the quantifications made by Zhao et al. (2022). Zhao et al. (2022) revealed that compound hot-dry events significantly influenced U.S. winter yield variability, causing a 2.2 ~ 3.0 % yield loss per standard unit. Additionally, our findings indicated that CWD significantly dominated China winter yield variability, leading to a 0.54 ~ 1.75 % yield loss per standard unit. Some studies have examined the influence of extreme wet and cold temperatures on crop yields (Wang et al., 2021). These investigations mainly utilize short-term controlled trials in the field, potentially facing spatial limitations in their assessments. Despite a declining trend in CWD in the China wheat belt, this decrease lacks statistical significance. Consequently, given the considerable sensitivity of both yield and grain numbers to CWD, it is crucial to investigate its impact on regional yields under future climate change.

Our modeling, utilizing aggregated data from diverse varieties and planting dates, uncovered variations in yield sensitivity to extreme climate indices across different varieties and sowing dates. We noted a less impact of extreme climate indices on wheat yield loss in early-maturing varieties compared to others. This could be attributed to the role that early-maturing varieties play in minimizing the probability of wheat exposure to extreme climate events (Collins and Chenu, 2021; Trnka et al., 2014). Collins and Chenu (2021) reported that sowing early-maturing varieties could enhance yields in the Australian wheat belt by decreasing the frequency of severe frost, heat, and drought stresses under climate change. Additionally, we found that the impact of  $HDD_{HD-MT}$  on yield loss showed no significant differences among the varieties (Fig. 6c). This lack of distinction could be attributed to the growing days that vary among these varieties, and there is not a significant distinction in heading date, with  $HDD_{HD-MT}$  potentially occurring around the heading stage. Regarding sowing date, the sensitivity of yield loss to extreme climatic indices decreased with a delayed sowing date (Fig. 6c). This is likely because extending the sowing period allows the reproductive growth phase of wheat to coincide with local summer rains, mitigating the negative impact of dry heat on wheat yields. However, the ability to adjust sowing schedules to minimize adverse climate effects depends on the crop type and local climate (Hunt et al., 2019; Minoli et al., 2022; Nouri et al., 2017). In the southern

hemisphere, early planting is more likely to counteract the effects of unfavorable climate on wheat yields (Hunt et al., 2019). However, there is significant uncertainty in sowing date adjustments in the northern hemisphere. Nouri et al. (2017) reported that delayed sowing dates could partially compensate for the detrimental effects of climate change-induced drought on rain-fed wheat yields in western and northwestern Iran. However, Jiang et al. (2023) suggested that early sowing was more resilient to future climate changes than late sowing for winter wheat at three sites in China. This contradiction may be attributed to regional differences, including climate and wheat varieties, in the study areas.

There are certain limitations in quantifying yield loss in response to extreme climate indices. Our study relies on field observations of winter wheat yield traits (1981–2010) from the China Meteorological Administration's nationwide agrometeorological observation network. However, the true extent of wheat exposure to compound cold-wet (CWD) and hot-dry (HDD) events might be underestimated because of the restricted number of observation sites and years. This raises the possibility that the model has underestimated yield losses due to CWD and HDD. Furthermore, since the observations mainly concentrated on late-growing traits of wheat, there is a shortage of data regarding biomass, leaf area index, and tiller number. Therefore, the inter-varietal differences in the sensitivity of yield traits of different wheat varieties to extreme climate indices may be underestimated. Lastly, given that our model is a statistical one, it lacks intrinsic mechanistic properties and cannot be extrapolated across temporal and spatial. Therefore, to predict the impacts of extreme temperatures and atmospheric humidity on agricultural production under climate change need to improve models mechanistic and the collaboration of different modelers (data-driven, mechanism-driven, and hybrid model-driven).

## 5. Conclusions

Based on the observed traits of winter wheat at 85 locations managed by the China Meteorological Center from 1981 to 2010, we quantified the effects of extreme compound events on various traits in winter wheat through a linear mixed modeling approach. The main conclusions drawn from the study were as follows:

- (1) Increasing trend of compound dry and hot days (1981–2010) dominates yield loss of winter wheat in China.
- (2) Compound wet and cold and dry heat have comparable negative effects on winter wheat in China.
- (3) Extreme weather has the lowest negative effect on wheat yield for early maturing varieties and late sowing measures.

## CRedit authorship contribution statement

**Tengcong Jiang:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Funding acquisition, Formal analysis, Data curation. **Liang He:** Data curation. **Hao Feng:** Funding acquisition. **Jianqiang He:** Writing – review & editing. **Qiang Yu:** Writing – original draft, Supervision, Funding acquisition, Data curation.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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## Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.agrformet.2024.110354.

## Data availability

Data will be made available on request.

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